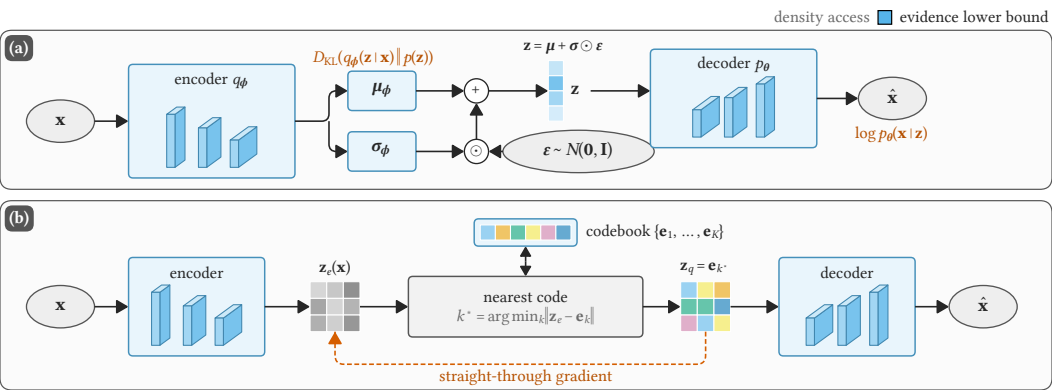
density access
evidence lower bound
(a)
x
encoder q_ϕ
$D_{KL}(q_\phi(\mathbf{z}|\mathbf{x}) \| p(\mathbf{z}))$
μ_ϕ
σ_ϕ
+
⊙
$\mathbf{z} = \mu + \sigma \odot \varepsilon$
z
$\varepsilon \sim N(\mathbf{0}, \mathbf{I})$
decoder p_θ
$\hat{\mathbf{x}}$
$\log p_\theta(\mathbf{x}|\mathbf{z})$
(b)
x
encoder
$\mathbf{z}_e(\mathbf{x})$
codebook $\{\mathbf{e}_1, \ldots, \mathbf{e}_K\}$
nearest code
$k^* = \arg\min_k \|\mathbf{z}_e - \mathbf{e}_k\|$
$\mathbf{z}_q = \mathbf{e}_{k^*}$
decoder
$\hat{\mathbf{x}}$
straight-through gradient